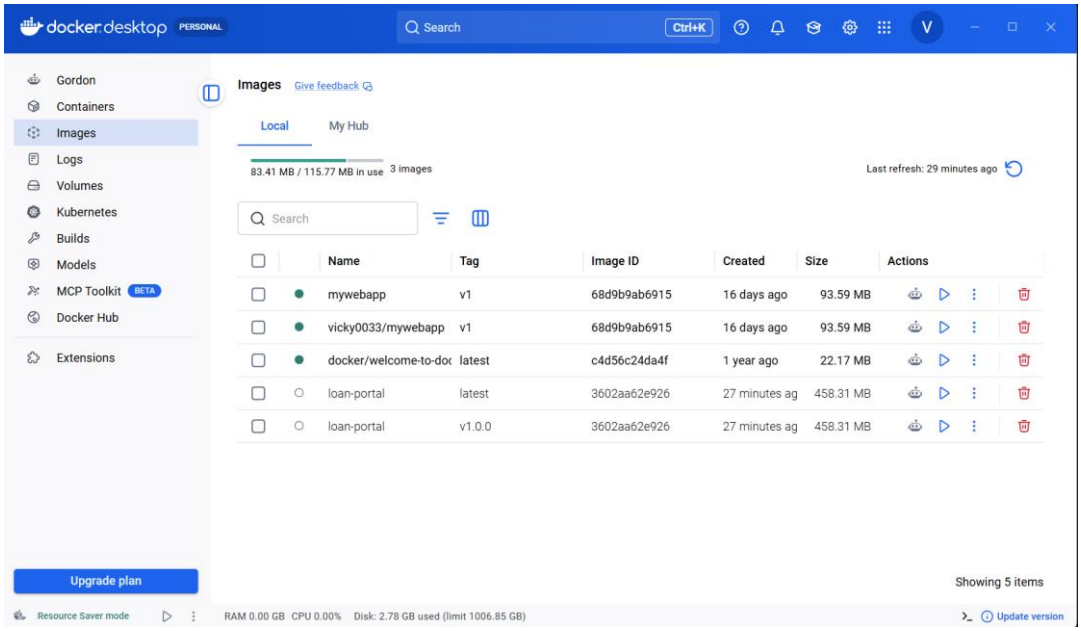
Semester	B.E. Semester VII – Computer Engineering
Subject	DevOps Lab
Subject In-charge	Dr. Snehal Andhare

Student Name	Vicky Pukale
Roll Number	23102A0033
Grade and Subject Teacher's Signature	

Task Number	12
Project Title:	Secure CI/CD Pipeline for a Loan Application Tracking Portal
Task Title	Jenkins-Docker Continuous Deployment: Extend the Loan Application Tracking Portal Jenkins pipeline to build a versioned Docker image, publish it to Docker Hub or a local registry and deploy a fresh container automatically after successful tests.
Deliverable:	**Upgrading `Jenkinsfile` for Docker Continuous Deployment**

Step-by-Step Jenkins-Docker CD Execution

1: Verify Docker Desktop is Running



2: Commit and Push the Updated `Jenkinsfile`

git add Jenkinsfile

git commit -m "ci(cd): integrate automated Docker image versioning and continuous container deployment"

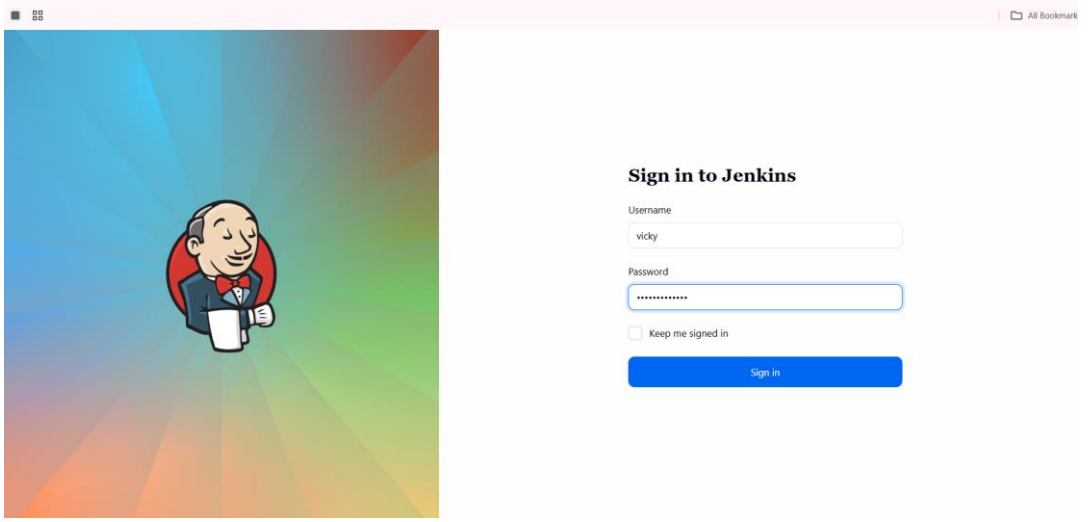
git push origin main

```
D:\Vicky_Project\DevOps\DevOps_Project>git add Jenkinsfile
warning: in the working copy of 'Jenkinsfile', LF will be replaced by CRLF the next time Git touches it

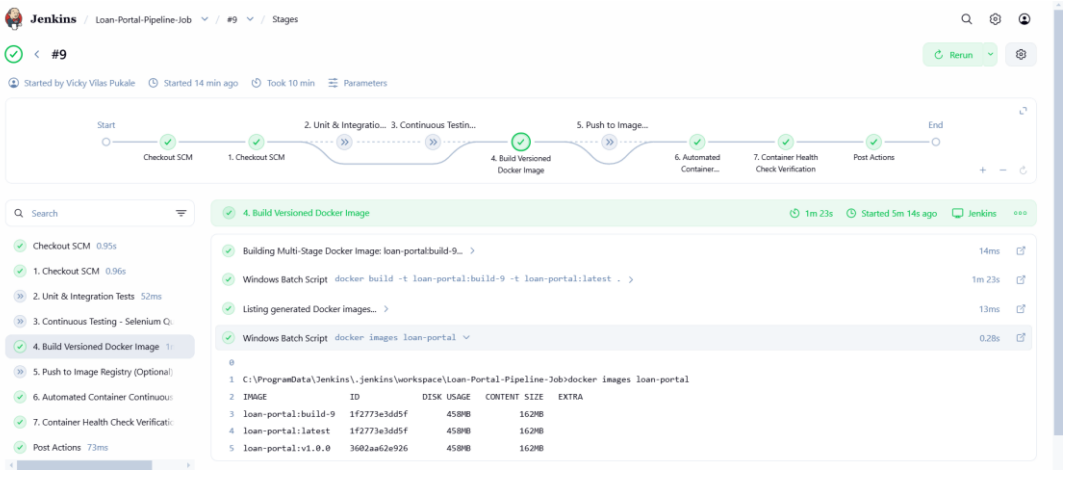
D:\Vicky_Project\DevOps\DevOps_Project>git commit -m "ci(cd): integrate automated Docker image versioning and continuous container deployment"
[main bc27765] ci(cd): integrate automated Docker image versioning and continuous container deployment
 1 file changed, 45 insertions(+), 23 deletions(-)

D:\Vicky_Project\DevOps\DevOps_Project>git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 1.31 KiB | 1.31 MiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/vickypukale/loan-application-tracking-portal.git
   0feb301..bc27765  main -> main
```

3: Trigger the Pipeline in Jenkins



4: Observe the End-to-End Pipeline in Stage View



```
[Pipeline] echo
🎉 END-TO-END DOCKER CONTINUOUS DEPLOYMENT SUCCESSFUL!
[Pipeline] echo
Deployed Image     : loan-portal:build-9
[Pipeline] echo
Container Name     : loan-portal-app
[Pipeline] echo
Target Environment : DEV
[Pipeline] echo
Live Portal URL    : http://localhost:8081/dashboard
[Pipeline] echo
============================================================
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Pretty-print ☐

{"status":"UP","components":{"db":{"status":"UP","details":{"database":"H2","validationQuery":"isValid()"}},"diskSpace":{"status":"UP","details":{"total":1081101176832,"free":1022611550208,"threshold":10485760,"path":"/app/.","exists":true}},"ping":{"status":"UP"}}}

5: Verify the Live Container in Terminal & Browser

